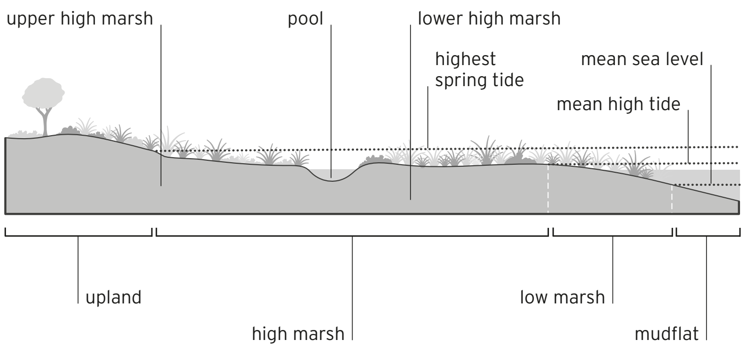
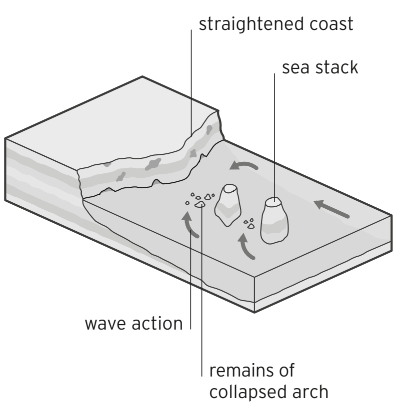
SALT MARSH

Coastal wetlands that are flooded and drained by saltwater brought in by tides are called salt marshes. Salt-tolerant vegetation grows in mud and peat, which may be deep and characterized by very low oxygen levels. Many areas of salt marsh are bordered by tidal flats. See also *tidal flat*.



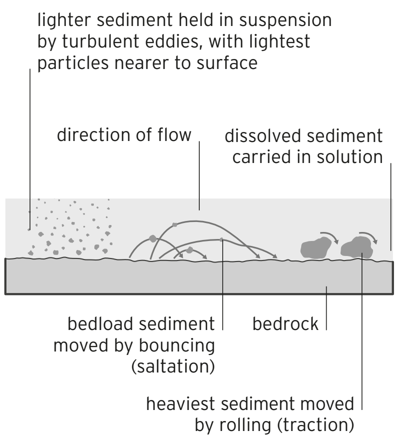
SEA STACK ▼ An offshore column of rock remaining after the erosion of adjacent sea cliffs, which are often of less-resistant rock.



SEA-ICE Frozen ocean water. The sea freezes in several stages, first grease ice, then pancake ice, before forming continuous sheets of sea-ice. See also *grease ice, pancake ice*.

SEAFLOOR SPREADING A process that occurs at mid-ocean ridges where new oceanic crust is formed and gradually moves away from the ridge. See *mid-ocean ridge* (panel), *spreading ridge, tectonic plate*.

SEAMOUNT An undersea mountain, usually volcanic in origin, whose peak does not reach the surface. See also *guyot*.



SEDIMENT ▲ Mud, sand, gravel, and other particles that have been transported by water, wind, gravity, or volcanic processes and later deposited. Moving water can transport its sediment load by several different processes.

SEDIMENTARY ROCK Material formed when mud, sand, gravel, or other particles are deposited in an ocean or lake, or on land, and are then hardened by the processes of compaction and cementation.

SEIF DUNE See *linear dune*.

SEISMIC WAVE A shock wave generated by an earthquake. There are three types of seismic wave: P-waves, S-waves, and surface waves.

SEMIDESERT An arid region that receives enough rainfall to support some plant life.

SERAC A pinnacle, or ridge, of glacial ice.

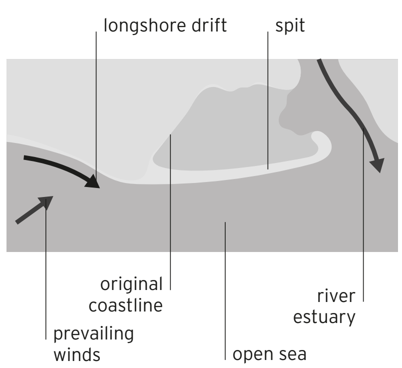
SHEET LIGHTNING An electrical discharge between two charged regions of one storm cloud. All or part of the actual discharge is obscured by cloud. It is also called intracloud lightning. See also *forked lightning, thunderstorm*.

SHIELD A large area of continental crust with ancient metamorphic rocks at the surface.

SHIELD VOLCANO The largest volcano type in terms of area, but with gently sloping sides. Shield volcanoes are the least violent, built mostly from liquid lava flows. See also *volcano* (panel).

SILL A thin, sheetlike igneous intrusion that forms when igneous rock forces its way between layers of sedimentary rocks. Most sills are roughly horizontal. See also *batholith, dike, igneous intrusion* (panel).

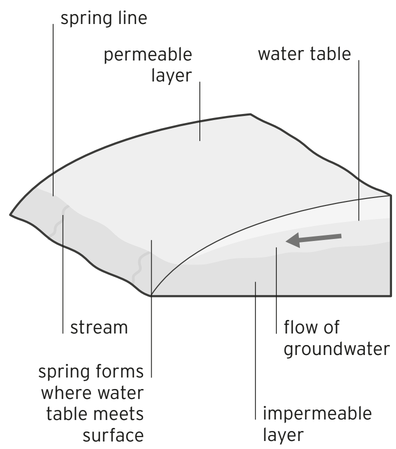
SINKHOLE A depression that forms where water has dissolved limestone bedrock. Sinkholes are characteristic of karst landscapes. See also *karst, limestone*.



SPIT ▲ A peninsula of sand or shingle connected to the shore at one end. Spits are usually created by longshore drift, often where the coastline changes direction abruptly. See also *bar, longshore drift, tombolo*.

SPREADING RIDGE The raised area on the ocean floor where two tectonic plates are slowly moving apart. Molten rock from the mantle wells up along a spreading ridge to solidify into new oceanic crust. Situated between the Pacific and Nazca Plates, the East Pacific Rise contains one of Earth's fastest spreading ridge systems. See also *boundary* (panel), *mid-ocean ridge* (panel), *seafloor spreading, tectonic plate*.

SPRING ▼ A surface outlet for water that has flowed from porous rocks underground. See also *groundwater*.



SPRING TIDE The tide with the greatest difference between low and high water. Spring tides occur twice a month when the gravitational pull on Earth's oceans of the Moon and Sun are working with each other. It falls much lower than a neap tide, exposing more of the shore, as well as rising much higher a few hours later. See also *neap tide*.

SPUR A sloping ridge projecting from the side of a mountain or valley. A truncated spur has a blunt end as a result of erosion by a glacier or faulting.